VINEETH GOLLA

AI/ML ENGINEER | MACHINE LEARNING ENGINEER | SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

AI/ML Engineer with 5+ years of software engineering experience spanning high-volume financial systems at JPMorgan Chase and large-scale AI/ML engineering at Google DeepMind. Built and optimized backend services, event-driven processing systems, distributed data pipelines, and production infrastructure before transitioning into AI/ML systems supporting model evaluation, inference, and scalable machine learning workflows. Strong expertise in Python, C++, Java, SQL, distributed systems, Kafka, PyTorch, Kubernetes, Docker, and cloud-based infrastructure, with a track record of improving system performance, automating engineering workflows, strengthening production reliability, and translating complex ML requirements into scalable production software.

PROFESSIONAL EXPERIENCE

AI/ML Engineer **June 2025 – Present | USA**
Google DeepMind

- Optimized AI/ML inference services and backend APIs by restructuring data-access paths, introducing targeted caching, and reducing service-to-service overhead, cutting average request latency by approximately 18% across targeted workloads.
- Scaled distributed processing pipelines for large AI datasets and model evaluation workloads, increasing batch-processing throughput by approximately 22% while maintaining validation, fault-handling, and data-integrity controls.
- Translated model-integration requirements from ML researchers and infrastructure teams into production-ready software components, resolving interface, scalability, and reliability constraints across 6+ AI/ML workflows.
- Engineered automated evaluation and validation utilities for AI outputs, enabling engineering teams to compare model behavior across test datasets, detect regressions, and validate model-related changes before production releases.
- Instrumented AI/ML services with structured logging, application metrics, distributed tracing, and failure alerts, reducing mean time to isolate recurring production issues by approximately 25%.
- Tuned containerized AI/ML workloads and Kubernetes deployment configurations by analyzing resource consumption, service limits, and workload behavior, reducing deployment-related failures by approximately 15%.
- Expanded automated unit, integration, and regression testing across AI-related services, increasing automated test coverage by approximately 20% and reducing defects reaching later validation stages.
- Streamlined CI/CD workflows for AI/ML services by automating build validation, test execution, deployment checks, and release steps, reducing manual release activities by approximately 30%.
- Resolved performance, scalability, and integration bottlenecks through architecture reviews, code analysis, and cross-functional debugging with software, ML, infrastructure, and research engineering teams.
- Hardened production AI/ML services through failure handling, input validation, service health checks, and recovery mechanisms, improving operational reliability across recurring model-serving and evaluation workflows.

Software Engineer **March 2020 – Dec 2023 | India**
JPMorgan Chase

- Modernized legacy transaction-processing components into modular backend services, reducing targeted processing latency by approximately 20% while simplifying service dependencies and improving maintainability.
- Re-engineered high-volume transaction workflows around event-driven processing, increasing peak processing throughput by approximately 25% while preserving validation, error-handling, and transaction-integrity controls.
- Optimized SQL execution paths across transaction-processing workloads through query restructuring, indexing improvements, and database-access tuning, reducing database response time by approximately 18%.
- Engineered Java and Python backend services and REST APIs for financial transaction workflows, implementing validation, exception handling, and service-level controls for high-volume production systems.
- Automated unit and integration testing across backend services, increasing test coverage by approximately 22% and reducing regression defects during recurring application releases.
- Containerized application services with Docker and supported Kubernetes-based deployments during cloud modernization initiatives, reducing environment-specific release issues by approximately 15%.
- Streamlined CI/CD pipelines for build, testing, validation, and deployment workflows, reducing manual deployment effort by approximately 30% and improving release consistency across environments.
- Strengthened production observability through structured logging, application monitoring, health checks, and failure-handling mechanisms, reducing time spent diagnosing recurring service incidents.
- Translated financial technology requirements into scalable software solutions by working with product managers, architects, QA engineers, database teams, and security stakeholders across multiple engineering workstreams.
- Diagnosed production failures, performance bottlenecks, and recurring application defects through log analysis, database investigation, and service-level debugging, contributing to approximately 15% fewer recurring issues across targeted services.

TECHNICAL SKILLS

Programming & Software Engineering: Python, C++, Java, SQL, Data Structures & Algorithms, Object-Oriented Programming, Multithreading, Async Programming

Machine Learning & Deep Learning: Machine Learning, Deep Learning, PyTorch, TensorFlow, JAX, Keras, Scikit-learn, Model Training, Model Evaluation, Model Optimization, Feature Engineering

Generative AI & LLMs: Generative AI, Large Language Models (LLMs), Transformers, LLM Evaluation, Prompt Engineering, RAG, Embeddings, Vector Search, AI/ML Evaluation, Responsible AI

AI/ML Engineering: ML Pipelines, Model Serving, Model Inference, ML Infrastructure, Distributed ML Systems, Data Processing, AI Model Integration, Performance Optimization, Experimentation, Model Monitoring

Data Engineering & Distributed Systems: Apache Kafka, PySpark, Pandas, NumPy, ETL/ELT, Data Pipelines, Distributed Computing, Event-Driven Architecture, REST APIs, Microservices, PostgreSQL, MySQL, Redis

Cloud, MLOps & Infrastructure: Google Cloud Platform (GCP), AWS, Docker, Kubernetes, Terraform, MLflow, CI/CD, GitHub Actions, Jenkins, Model Deployment, Infrastructure as Code, Cloud Monitoring

AI Frameworks & Model Serving: PyTorch, JAX, TensorFlow, Keras, Hugging Face Transformers, ONNX, vLLM, Model Serving, Batch Inference, Real-Time Inference

Testing, Observability & Development: Pytest, Unit Testing, Integration Testing, Regression Testing, Distributed Tracing, Structured Logging, Performance Monitoring, Debugging, Git, GitHub

Engineering Practices: System Design, Scalable Architecture, Software Design Patterns, Agile/Scrum, Code Reviews, Technical Documentation, Cross-Functional Collaboration, Production Support

EDUCATION

Saint Louis University **St. Louis, Missouri**
Master's in Information Systems